WASTEWISE-ML · DESIGN DECISION RECORD

WasteWise-ml TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	WML-DDR-001	Version	0.3
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/WasteWise-ml		
Source	docs/decisions/0001-trl2-review-decisions.md @ 9031e25		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 and O2 decided by Amish on 2026-10-02 as recommended (SWaCH in Pune as first candidate partner; MIT-only contributing guide)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review points

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items D1 to D8; items O3 and N1 to N3 decided by Amish on 2026-09-25 through WML-DDR-002); items O1 and O2 decided on 2026-10-02 as recommended (Amish, 2026-10-02: "i approve your recommendations for all 555 open decisions.")

Context

The TRL 2 review note (`docs/REVIEW.md`, session of 2026-09-25, "/populate to a strong TRL 2") listed ten items marked "Proposed, awaiting Amish". On 2026-09-25 Amish reviewed the review points for every repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." Every item that carried a recommendation is therefore decided as recommended. Items without a recommendation stay open.

Options considered

Table 1. Items with a recommendation.

#	Item	Options	Recommendation
D1	Task form	A: single item on a mat, image classification; B: multi-item detection in a pile	A for TRL 3; B as a later extension
D2	Backbone and runtime	MobileNetV3-Large, EfficientNet-Lite0 or a small vision transformer; TensorFlow Lite, ONNX	MobileNetV3-Large; TensorFlow Lite as the main export, ONNX as a second export
D3	Taxonomy changes	Add or leave out the `composite` class and the `unsure` and `hazard` outputs	Keep them
D4	Grade mapping	Per-site configuration file; site-specific models	Configuration file per site
D5	Scan mode	Tap-to-scan; live camera preview	Tap-to-scan, for battery life
D6	Data ownership and consent	Field data co-owned by the pickers' organization, opt-in upload, no faces; project-owned data	Co-owned, opt-in, no faces, written into a protocol with the partner before any collection
D7	Pay rate for labeling	Fixed planning rate; rate set with the partner	Set with the partner at or above the local living wage
D8	Pilot phones	Reference phone per station; pickers' own phones	One reference phone per station for the evaluation; support own phones in use

Decision

- **D1.** Decided by Amish, 2026-09-25: go with recommendation. WasteWise-ml classifies one item at a time on a light mat under a phone on a stand (handheld use also supported). Pile detection is a later extension, not part of this design.

- **D2.** Decided by Amish, 2026-09-25: go with recommendation. MobileNetV3-Large backbone with a class head and a grade head; 8-bit TensorFlow Lite as the main export and ONNX as a second export. EfficientNet-Lite0 stays a comparison candidate for later work.
- **D3.** Decided by Amish, 2026-09-25: go with recommendation. `ml/data/taxonomy.yaml` keeps the `composite` class and the `unsure` and `hazard` outputs.
- **D4.** Decided by Amish, 2026-09-25: go with recommendation. Buyer grade names, bin numbers, colors and voice prompts are set per site in a configuration file.
- **D5.** Decided by Amish, 2026-09-25: go with recommendation. The app uses tap-to-scan; the camera does not run between scans. WML-CAL-001 adds that the screen must also sleep between scans to meet R10 without the power bank.
- **D6.** Decided by Amish, 2026-09-25: go with recommendation. Field photos are collected only with informed consent, contain no faces, are uploaded only on opt-in, and are co-owned by the pickers' organization. The protocol is written with the partner before any collection.
- **D7.** Decided by Amish, 2026-09-25: go with recommendation. The labeling pay rate is set with the partner at or above the local living wage. The $6 per hour in `bom/bom.csv` remains a planning figure only.
- **D8.** Decided by Amish, 2026-09-25: go with recommendation. Each pilot station gets one reference phone (BOM line 1) for the evaluation; pickers' own phones are supported in use.

Budget and pitch: this is a software repository, licensed MIT only, and `budget_usd` stays null. The TRL 2 review recommended no pitch or problem change ("Pitch. Unchanged."), so the pitch and problem lines in `project.yaml` and `README.md` are unchanged. The kit's blueprint sheet label stays overridden to MIT for this repository.

Items listed as open at TRL 3 (O1 and O2 carry no recommendation and stay "Proposed, awaiting Amish"; O3 carried a suggestion and is now decided, see WML-DDR-002):

- **O1.** First partner organization, city and buyers for co-design and the field set. No recommendation; portfolio guidance is that community designs pick co-design partners per area later. Decided by Amish on 2026-10-02 as recommended in WML-DEC-001: a member-owned waste picker cooperative that already sorts dry waste for scrap buyers, with SWaCH in Pune, India, as the first candidate to approach, its scrap dealers as the buyers and Pune as the city.
- **O2.** `CONTRIBUTING.md` still says hardware contributions are under CERN-OHL-S v2, although this repository is MIT only. The review noted that Amish may want to remove the line but made no recommendation. Decided by Amish on 2026-10-02 as recommended in WML-DEC-001: the line is removed and all contributions are licensed under MIT.
- **O3.** A kit option for software repositories so that drawing sheets default to MIT (the review suggested it; it changes `.kit/`, which this repository does not own). Decided by Amish, 2026-09-25: go with recommendation. Recorded as a cross-repo action for the kit owner; `.kit/` is not edited here.

New items raised at TRL 3 (decided through WML-DDR-002):

- **N1.** Hazard evaluation size (R3). WML-CAL-001 shows that 200 hazard items prove 98 % recall only if the model misses none. Options: keep 200; raise to 400 (proves 98 % at 95 % confidence with up to 3 misses, an 80 % chance of passing if true recall is 99.5 %); raise to about 1,000 (80 % chance of passing if true recall is 99.0 %). Recommendation: 400, collected by targeted sampling. Decided by Amish, 2026-09-25: go with recommendation.

- **N2.** ZeroWaste is licensed CC BY-NC 4.0, which does not fit MIT model weights. Options: exclude it from training and use it only for non-commercial research comparisons; exclude it entirely; ask the authors for other terms. Recommendation: exclude it from training. Decided by Amish, 2026-09-25: go with recommendation.
- **N3.** Camera height. The TRL 2 concept put the phone about 470 mm above the mat; at that height a typical 26 mm-equivalent phone camera sees about 626 x 469 mm, wider than the 450 mm mat. Options: lower the lens to 440 mm (frame 586 x 439 mm, used in the model); keep 470 mm and widen the mat to 500 mm. Recommendation: 440 mm. Decided by Amish, 2026-09-25: go with recommendation.

Consequences

- WML-PRB-001, WML-PRC-001 and WML-REQ-001 move to version 0.3 and state D1 to D8 as decisions rather than proposals.
- WML-CAL-001 sizes the dataset, model, energy and evaluation on these decisions.
- The build123d station model (`cad/src/model.py`) and drawing WML-DWG-001 use the 440 mm lens height of N3, now decided (WML-DDR-002).
- Nothing here starts TRL 4 work: no training, data collection, app code or field trial.